# SANJANA FERNANDO

Full Stack Software Engineer (AI/ML Focused)

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## PROFESSIONAL SUMMARY

Fresh Computer Engineering graduate (BSc, Sep 2026) specializing in applied machine learning – reinforcement learning, computer vision, and LLM-based multi-agent systems – with production full-stack experience (Next.js, Node.js) shipping and integrating ML-driven features end-to-end. Reduced page load times on a live full-stack platform during a professional internship.

## TECHNICAL SKILLS

- **Machine Learning & AI:** Python, PyTorch, Scikit-learn, Reinforcement Learning (DDQN), Computer Vision, MediaPipe, OpenCV, LangChain, LangGraph, RAG, Google Gemini, Pandas, NumPy
- **Web Development:** Next.js, React.js, Node.js, Express.js, TypeScript, Vite.js, REST APIs
- **Databases:** PostgreSQL, Supabase (pgvector), MongoDB, MySQL
- **Mobile Development:** Flutter, Dart, Firebase
- **Security & Tools:** Git, Web Security Models, Cryptography (Encryption/Hashing)

## PROFESSIONAL EXPERIENCE

**Software Engineering Intern** *May 2025 – Feb 2026*
*BotCalm (PVT) Ltd.* *Sri Lanka*

- Collaborated in an agile team to build a casino fraud detection system using Next.js, engineering anomaly-detection logic and secure data pipelines to identify and flag suspicious user behavior in real time.
- Co-engineered a full-stack luxury marketplace platform using Next.js, implementing 10+ core front-end UI components and optimizing back-end API logic to reduce page load times by 15%.
- Developed high-performance web applications using Vite.js, integrating complex UI animations with Framer Motion to deliver fluid, highly interactive front-end user interfaces.

**Bank Trainee**, Peoples Bank, Sri Lanka *Dec 2021 – Feb 2022*
Processed 150+ daily financial transactions with 100% cash-balancing accuracy; handled customer issues under pressure per strict banking compliance standards.

## KEY PROJECTS

**Hospital Bed Allocation & Management Platform – Reinforcement Learning** *GitHub / Live*

- Designed and trained a custom Double DQN (DDQN) agent in PyTorch to optimize hospital bed allocation, deployed behind a Next.js dashboard with a Python backend to maximize asset utilization and reduce simulated patient wait times.
- **Technologies:** Next.js, React.js, Tailwind CSS, Python, PyTorch, DoubleDQN (DDQN), NumPy, Pandas, SciPy

**BidPilot – AI-Powered RFP & Tender Proposal Platform** *GitHub / Live*

- Built a LangGraph multi-agent RAG workflow (Python, FastAPI) to automate RFP document parsing, semantic knowledge retrieval, and compliance-checked bid generation, paired with a Next.js dashboard for end-to-end proposal automation.
- **Technologies:** Python, FastAPI, LangGraph, LangChain, Google Gemini, Supabase (PostgreSQL/pgvector), PDFPlumber, Next.js, TypeScript

**AI-Powered Hand Gesture Virtual Mouse** *GitHub*

- Engineered a real-time computer vision system using MediaPipe and OpenCV to track hand landmarks and classify gestures via a PyTorch neural network for hands-free system control.
- **Technologies:** Python, PyTorch, MediaPipe, OpenCV, Deep Learning

**BookTalk – AI Chat Application** *GitHub*

- Built a Next.js chat application with a responsive web interface and streaming LLM responses via a Groq-hosted Llama model.
- **Technologies:** Next.js, React.js, TypeScript, Tailwind CSS, Vercel AI SDK, Groq, OpenAI SDK

**Additional Projects:**

- **Secure File Drop** – Full-stack file transfer app with authentication and access controls. GitHub

- **E-Chess** – Full-stack platform for real-time competitive chess tournaments. GitHub
- **Fitness Tracking App** – Flutter/Firebase mobile app for meal plans and workout tracking. GitHub

## EDUCATION

**Bachelor of Science in Computer Engineering Specialization** **Graduated: Sep 2026**
University of Ruhuna | Department of Electrical and Information Engineering
**Final CGPA: 3.33 / 4.00**

- **Relevant Coursework:** Artificial Intelligence & Machine Learning, Data Structures & Algorithms (DSA), Web Development, Object-Oriented Programming (OOP), UI/UX, Project Management, Networking, Analog & Digital Electronics

**GCE Advanced Level (2020)**
Physics – A, Combined Maths – A, Chemistry – B *Z-score: 1.9*

## CERTIFICATIONS

- Python for Data Science and Machine Learning Bootcamp – Udemy (2023)
- The Ultimate React Course: React, Next.js, Redux & More – Udemy (2023)
- React, NodeJS, Express & MongoDB – The MERN Fullstack Guide – Udemy (2024)
- Flutter & Dart – The Complete Guide – Udemy (2024)

## EXTRA-CURRICULAR ACTIVITIES

- **IEEEXtreme 18.0 Hackathon (Logistics Lead):** Orchestrated operations and resource allocation, helping the university secure its highest-ever ranking.
- **EMINENCE 4.0 (Logistics Committee Member):** Streamlined event logistics, contributing to the Best Affinity Group Project Award in the IEEE Sri Lanka Section.

## REFERENCES

**Dr. Geeth Priyankara**
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